Requirement Document: N8N Hands-On Intermediate Course

Topic 6: Multi-Source Data Aggregation, PDF Invoice Extraction & Workflow Synchronization

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1. Document Control

Field	Value
Document Title	Topic 6: Multi-Source Data Aggregation & PDF Invoice Extraction - Requirement File
Course	N8N - Hands On Intermediate Course (Level 2)
Target Audience	Enterprise Automation Engineers, Data Engineers, Solutions Architects
Version	1.0
Status	Approved & Completed
Last Updated	October 16, 2026
Prepared By	Lead Automation Architect

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2. Learning Concepts & Objectives

2.1 Course Objective

Build an enterprise Customer 360 data synchronization and financial reconciliation pipeline in n8n. The workflow ingests customer master records from a CRM API, transaction history from a SQL database, and unstructured supplier expense invoices from binary PDF files. Using n8n's **Extract from File (PDF)** node, regular expression parsing, and multi-stage **Merge** nodes, it unifies heterogeneous data streams on `customerId`, computes financial reconciliation metrics (Revenue, Expenses, Net Margin, and Tier classification), and exports synchronized Customer 360 profiles to a cloud data warehouse mock endpoint.

2.2 Learning Scope

- **Multi-Branch Parallel Data Ingestion:** Initiating 3 asynchronous ingestion branches from a single `Manual Trigger` (`Code_MockCRMData`, `Code_MockSQLData`, `Code_SimulatePDFInvoice`).
- **Binary Document Generation & Parsing:** Synthesizing an in-memory binary PDF invoice containing supplier billing text and extracting raw textual strings using the `Extract from File (PDF)` node without external OCR licenses.
- **Regex Financial Line-Item Extraction:** Parsing unstructured invoice text using targeted regular expressions to extract `customerId` and numerical `pdfExpenseUSD`.
- **Multi-Stage Keyed Merging:** Leveraging two sequential `Merge` nodes (Combine mode by matching field `customerId`) to stitch CRM master accounts with SQL sales totals, and then with PDF vendor expenses.
- **Customer 360 Metric Computation:** Calculating net profit margin (`totalSpendUSD - pdfExpenseUSD`) and assigning strategic account tiers (`Enterprise Tier 1` vs `Standard`).
- **Warehouse Data Load & Audit Aggregation:** Dispatching unified records to an external data warehouse ingestion endpoint (`HTTP_PostToWarehouse`) and recording completion audit metrics (`Code_AuditSummary`).

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3. Workflow Architecture

```mermaid

flowchart TD

Trigger["Node 1: Manual_Trigger
Initiate Sync Run"]
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%% Three parallel branches

Trigger --> CRM["Node 2A: Code_MockCRMData
Fetch Profiles: CRM API"]

Trigger --> SQL["Node 2B: Code_MockSQLData
Query Orders: SQL DB"]

Trigger --> PDF["Node 2C: Code_SimulatePDFInvoice
Supplier Billing PDF Attachment"]

%% PDF Extraction Line

PDF --> Extract["Node 3: Extract_PDF_Invoice
Extract from File: PDF to Text"]

Extract --> Parse["Node 4: Transform_ParseInvoiceText
Regex Parser & Expense Extraction"]

%% First Merge

CRM --> Merge1{"Node 5: Merge_CRM_With_SQL
Combine by Key: customerId"}

SQL --> Merge1

%% Second Merge

Merge1 --> Merge2{"Node 6: Merge_With_PDFExpenses
Combine by Key: customerId"}

Parse --> Merge2

%% Transformation & Analytics

Merge2 --> C360["Node 7: Transform_ComputeCustomer360
Net Margin, Total Spend, Tier"]

%% Warehouse Export

C360 --> PostWH["Node 8: HTTP_PostToWarehouse
POST to Data Warehouse"]

PostWH --> Audit["Node 9: Code_AuditSummary
Output Synchronization Metrics"]

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style Merge1 fill:#fefce8,stroke:#ca8a04,stroke-width:2px

style Merge2 fill:#fefce8,stroke:#ca8a04,stroke-width:2px

style Extract fill:#fdf4ff,stroke:#9333ea,stroke-width:2px

style C360 fill:#f0fdf4,stroke:#16a34a,stroke-width:2px
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style Audit fill:#ecfdf5,stroke:#059669,stroke-width:2px
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4. Node Configuration Specifications

Node ID	Node Name	Node Type	Critical Parameters & Configuration
N6-01	Manual_Trigger	n8n-nodes-base.manualTrigger	Manually starts the 3-branch pipeline.
N6-02A	Code_MockCRMData	n8n-nodes-base.code (v2)	Emits CRM profiles (CUST-1001 Apex Logistics, CUST-1002 Nexus BioTech).
N6-02B	Code_MockSQLData	n8n-nodes-base.code (v2)	Emits SQL transactions (CUST-1001: $145,200.50, CUST-1002: $18,400.00).
N6-02C	Code_SimulatePDFInvoice	n8n-nodes-base.code (v2)	Generates valid binary PDF invoice buffer in base64 format under binary property data.
N6-03	Extract_PDF_Invoice	n8n-nodes-base.extractFromFile (v1)	Operation: pdf, Binary Property: data. Extracts text payload.
N6-04	Transform_ParseInvoiceText	n8n-nodes-base.code (v2)	Uses regex /CUSTOMER ID:\s*(CUST-[0-9]+)/i and /EXPENSE AMOUNT:\s([0-9.]+)/i to output { customerId, pdfExpenseUSD }.

N6-05	Merge_CRM_With_SQL	n8n-nodes-base.merge (v2.1)	Mode: combine, Combination Mode: mergeByFields, Field 1: customerId, Field 2: customerId. Combines CRM + SQL.
N6-06	Merge_With_PDFExpenses	n8n-nodes-base.merge (v2.1)	Mode: combine, Combination Mode: mergeByFields, Field 1: customerId, Field 2: customerId. Stitches PDF expense line items.
N6-07	Transform_ComputeCustomer360	n8n-nodes-base.code (v2)	Computes netMarginUSD = rev - exp and assigns Tier (Enterprise Tier 1 if spend >= $100K else Standard).
N6-08	HTTP_PostToWarehouse	n8n-nodes-base.httpRequest (v4.2)	Method: POST, URL: https://httpbin.org/post, Body: ={{ JSON.stringify($json) }}. Dispatches payload to mock warehouse.
N6-09	Code_AuditSummary	n8n-nodes-base.code (v2)	Emits audit summary { warehouseSyncStatus: "SUCCESS", totalCustomerProfilesSynced: 2 }.

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5. Verification & Testing Checkpoints

5.1 PDF Extraction Validation

- Inspect Extract_PDF_Invoice output: verifies that text property contains CUSTOMER ID: CUST-1001 and EXPENSE AMOUNT: 12500.00 USD.
- Inspect Transform_ParseInvoiceText: confirms structured JSON { customerId: "CUST-1001", pdfExpenseUSD: 12500 }.

5.2 Multi-Stage Keyed Merge Verification

- Merge_CRM_With_SQL `accurately pairs CRM profiles with SQL total spends on` customerId.
- Merge_With_PDFExpenses `attaches the $12,500 PDF expense to` CUST-1001 `while maintaining` CUST-1002 `with 0 expense.`

5.3 Financial Reconciliation Output

- CUST-1001`: Spend = $145,200.50, Expense = $12,500.00, Net Margin = $132,700.50, Tier =` Enterprise Tier 1.
- CUST-1002`: Spend = $18,400.00, Expense = $0.00, Net Margin = $18,400.00, Tier =` Standard.
- Code_AuditSummary `confirms` totalCustomerProfilesSynced: 2`.